ICIF-Net: Intra-Scale Cross-Interaction and Inter-Scale Feature Fusion Network for Bitemporal Remote Sensing Images Change Detection

Yuchao Feng, Honghui Xu, Jiawei Jiang, Hao Liu, and Jianwei Zheng

***Abstract*— Change detection (CD) of remote sensing (RS) images has enjoyed remarkable success by virtue of convolutional neural networks (CNNs) with promising discriminative capabilities. However, CNNs lack the capability of modeling long-range dependencies in bitemporal image pairs, resulting in inferior identifiability against the same semantic targets yet with varying features. The recently thriving Transformer, on the contrary, is warranted, for practice, with global receptive fields. To jointly harvest the local-global features and circumvent the misalignment issues caused by step-by-step downsampling operations in traditional backbone networks, we propose an intra-scale cross-interaction and inter-scale feature fusion network (ICIF-Net), explicitly tapping the potential of integrating CNN and Transformer. In particular, the local features and global features, respectively, extracted by CNN and Transformer, are interactively communicated at the same spatial resolution using a linearized Conv Attention module, which motivates the counterpart to glimpse the representation of another branch while preserving its own features. In addition, with the introduction of two attention-based inter-scale fusion schemes, including mask-based aggregation and spatial alignment (SA), information integration is enforced at different resolutions. Finally, the integrated features are fed into a conventional change prediction head to generate the output. Extensive experiments conducted on four CD datasets of bitemporal (RS) images demonstrate that our ICIF-Net surpasses the other state-of-the-art (SOTA) approaches.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), inter-scale feature fusion, intra-scale cross-interaction, remote sensing (RS), Transformer.**

I. Introduction

CHANGE detection (CD) of bitemporal remote sensing (RS) images, as an increasingly important part of understanding land surface changes, is widely applied in urban expansion surveys [1], forest cover mapping [2], disaster damage assessment (e.g., landslide, earthquake, and so on) [3], [4], and other aspects. While the exact definition of CD may vary across applications, the primary goal is to assign binary bales on the paired pixels of two co-registered images taken at different periods. With the rapid development of optical sensors, the automatic CD technology has raised increasing attention, which can reduce massive labor costs and time consumption.

Although the availability of high-resolution optical images is opening up new avenues for blooming RS applications, CD at a fine scale remains a challenging job for three respects. First, the differences in imaging conditions caused by shooting at different periods, including lighting changes, season alternatives, and appearance variations, result in varied colors and shapes for the same semantic object in given image pairs. Second, the size of the targeted regions varies drastically, incurring a deeper consideration of smooth contours. Third, the interference of irrelevant changes inevitably leads to negative impact on the real change features. All these challenges give rise to distinct features both spatially and temporally for the regions with the same semantic concept. Technically speaking, a robust CD approach is considered to be able of: 1) discovering high-level semantic change of the region of interest and 2) capturing the intrinsic characteristics both locally and globally.

The traditional CD technique can be generally categorized into three branches: arithmetical-based, transformation-based, and post-classification methods. Considering the explicit discovery of land cover changes in a supervised manner, a method of extracting the changed areas after independent classification of image pairs [5] was proposed to guarantee an acceptable detection performance. However, although the post-classification method bypasses the difficulty of detecting changes from the original images, it is highly sensitive to the results of pre-learned features. Arithmetic operations, such as image difference [6], image regression [7], and image quantification [8], generate a difference map by directly comparing the pixel values in the bitemporal images and then distinguish the changing areas from the constant ones using a certain thresholding technique. Traditional machine learning methods, such as support vector machine [9], K-nearest neighbor (K-NN) [10], and random forest regression [11], can transform image domains to achieve heterogeneous CD. As a pixel-based analysis ignores spatial contextual information, object-based CD methods [12] are studied, which extract features from segmented image patches and identify changes in an object-based manner. Unfortunately, both arithmetical-based

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and transformation-based methods follow an unsupervised learning scheme, which is highly dependent on the empirical design and performs poorly on high-resolution images. Recently, the performance of unsupervised CD methods that eliminate the exorbitant and arduous labeling costs has been significantly boosted by virtue of the studies, such as the robust adaptive K-NN graph representations [13], the structure consistency-based graph construction [14], and the dynamic pseudo-label generation mechanism [15]. Yet, the supervised approaches still dominate the mainstream due to their highly appealing performance.

With impressive nonlinear mapping capability, deep learning-based models can capture fine image details and complex texture features, pushing traditional models aside in CD tasks. For instance, deep belief networks (DBNs) [16], stacked autoencoders (SAEs) [17], and convolutional neural networks (CNNs) [18], [19] generate difference images or change vectors of bitemporal images by extracting abstract depth features for comparison and then acquire the final change maps based on clustering or classification methods. However, both DBNs and SAEs belong to fully connected networks, which indicates that they suffer from a substantial loss of spatial information due to the restricted 1-D input form. CNNs [20] have shown remarkable ability in harvesting enriched and abstract local contextual features due to two well-known mechanisms of local connectivity and weight sharing. Following this line, a typical scheme is to modify an off-the-shelf model, e.g., the ones that may be originally developed for single-image semantic segmentation, to the CD task of bitemporal image pairs [21]. In [22], straightforward concatenation and convolution operations are performed to directly fuse the features extracted from the RS images for CD application. Unfortunately, the purely CNN-based model is inherently limited to the size of the reception field and is thirsty for the global dependencies. In terms of this issue, various applications of RS images, such as classification [23], target detection [24], and CD [25], employ spatial attention and channel attention to incorporate global concepts and guarantee a remarkable performance gain. In [25], the multilevel features of the bitemporal images and the difference features in-between the image pair are fused through the attention mechanism and further used for change map reconstruction. However, simply using attention to re-weight the fused features in the channel or spatial dimension is still struggling to relate long-range dependencies.

The Transformer model, being adept at capturing global concepts, has wowed the crowd for breaking through the limited reception field of CNNs. However, the outputs of Transformers at different stages are evenly uniform global representations, causing a great redundancy between shallow and deep layers [26]. In contrast, CNNs progressively enlarge the receptive field as the networks deepen, thus capturing features from subtlety to wholeness. To answer the question that how to effectively encode both local characteristics and global concepts, a straightforward idea is to integrate CNN and Transformer into a unified network. As demonstrated in [26], incorporating fine-grained information captured by CNN to the shallow layers of Transformer can boost the capabilities of data representation and contextual modeling. Following a sequential manner, several works [27]–[29] have also shown the benefits of combining CNN and Transformer, either employing convolutions at the initial stage or interleaving them into certain Transformer blocks. However, while promising detection performance can be achieved, the sequential treatment is still suboptimal, because it lacks the direct cross-interaction between CNN and Transformer.

To escape the limitations of separately or sequentially using CNN and Transformer, we propose a new model, namely, intra-scale cross-interaction and inter-scale feature fusion network (ICIF-Net), that extracts local multiscale features and global multiscale representations in a parallelized manner, followed by the acquirement of changing binary graph using intra-scale cross-interaction and inter-scale feature fusion. The contribution of our work can be summarized as follows.

1) To tap the potential of integrating CNN and Transformer, we shift the design paradigm from series to parallel and propose a dual-branch structure to couple the local and global features. By leveraging CNN's efficiency in extracting local features as well as Transformer's capability in global modeling, the parallel treatment jointly delivers the fine-grained details and the target-level interests for CD of RS image pairs.
2) With respect to the information communication between the separate branches and the aggregation of multiscale features, the intra-scale cross-interaction module and the inter-scale feature fusion module (FM) are developed for modeling the spatial–temporal contextual information and generating better representations of feature maps. Notably, feature extraction and fusion of bitemporal images can perform in parallel as well.
3) Extensive experiments on four typical datasets in bitemporal RS image CD demonstrate that the proposed ICIF-Net consistently outperforms other state-of-the-art (SOTA) methods. The trained model has been uploaded at https://github.com/ZhengJianwei2/ICIF-Net for evaluation, whose source code will be released soon.

The remainder of this article is organized as follows. In Section II, a brief review of related work is presented. The details of our proposed CD framework are described in Section III. Comprehensive experiments are conducted in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Work

The models for CD of bitemporal RS images can be archived from different views. Without consideration of semantic segmentation, CD relies solely on change labels can be generally divided into two main streams. One is the patch-based approach [30], which models CD as a similarity detection task on center prediction. Another is the pixel-level approach [21], [22], [25], [31], [32], which directly predicts the change map from the paired images and is considered to be more efficient and effective. From the view of when to fuse the bitemporal RS images, the detection methods can also be divided into two groups. One is the image-level method [22]

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that merges bitemporal images into one input and transmits the concatenated result to the semantic segmentation model, and the other is the feature-level method [21], [25], [31], [32] that aggregates the feature maps and further makes decisions on the fused results. As deep learning-based methods have achieved great success in the CD field, in the sequel, we roughly review the mostly related two branches, i.e., CNN-based and Transformer-based models, which are both the rooted studies of our work.

A. CNN-Based Model

To cope with the complexity of targets and the variation in imaging conditions, many recent studies aim to improve the generalization performance of neural networks by introducing strategies that can facilitate contextual information modeling, which is essential for identifying changes from RS image pairs. Among them, the multiscale feature fusion (MF) strategy [21], [25], [31] is second to none, which can provide both the low-level features that facilitate target localization and the high-level features with sufficient semantic information. To fully exploit the potential of multilevel features, Zhang *et al.* [25] directly and layer-wisely approximated the intermediate output to the spatially scaled change label in the decoder stage. Shi *et al.* [31] concatenated feature maps of different convolutional stages to fit the change labels and additionally bound the two max-scaled feature maps with different similarity measures, resulting in better feature representation. Furthermore, to increase the size of reception field, several schemes have been extensively studied and adopted, including deepened network architecture [21], dilated convolutional approaches [33], and attention mechanisms [21], [25], [31], [32]. Notably, attention mechanisms have evolved in various forms, such as channel attention [21], [34] that recalibrates channel-wise feature response and spatial attention [32] that highlights task-relevant spatial features. Especially in [25] and [31], channel attention and spatial attention are jointly considered to capture features with more discriminative properties.

Note again that the multiscale feature maps with local characteristics extracted by CNNs are of great significance for the CD of bitemporal RS image pairs. However, CNNs lack the ability to model long-range dependencies and global contextual interactions. Therefore, we conceive the proposal of incorporating CNNs as a feature extraction branch while introducing another branch that captures global features to achieve feature supplementation.

B. Transformer-Based Model

Since its creation, Transformers have been extensively employed in the field of natural language processing to model long-range dependencies with ease. With its remarkable modeling capabilities, Transformer exhibits considerable or even better performance as the convolutional counterpart, which has made a huge splash in the field of computer vision, including image classification [35], [36], semantic segmentation [36], [37], object detection [36], [38], image generation [39], video retrieval [40], and target tracking [41].

By emphasizing the importance of extracting global features, Transformer has been incredibly advanced in the RS community. For instance, in [42] and [43], the Transformer architecture is directly introduced into the classification task of RS images, generating amazing performance. To overcome the drawback that the fixed geometric structure of the convolutional kernel hinders the remote interaction between long-range features, Zhong *et al.* [44] proposed a spatial attention module and a spectral association module following a self-attentive mechanism. In [45], a joint CNN and Transformer method for the synchronous extraction of coarse-grained and fine-grained features is applied to semantic segmentation of RS images. Besides, Ma *et al.* [46] first introduced a Transformer to single synthetic aperture radar images for 3-D detection of oil tank targets.

With regard to the CD task of bitemporal RS images, Chen *et al.* [29] recently proposed a bitemporal image Transformer (BIT) module, which uses one Transformer encoder to model the spatial–temporal context based on the abstract feature maps from CNN. Subsequently, the learned context-rich tokens are fed back to the pixel space via two Transformer decoders. The sequential treatment in BIT is beneficial to efficiency guarantee, yet the direct information interaction between CNN and Transformer is ill-considered. Inspired by this, we parallelize the CNN and Transformer to achieve complementarity between local features and global features. In addition, to compensate for the multiscale features neglected by BIT, we focus on multiscale information utilization following a two-branch design.

III. Methodology

In this section, we present our CD framework termed ICIF-Net. As shown in Fig. 1, our method mainly consists of three stages. For any pair of images to be detected, we first introduce two subnetworks, CNN and Transformer, to parallelly obtain fine-grained local features and target-level global features, respectively. Moreover, to effectively fuse multiscale feature maps extracted from the two backbones, we propose an intra-scale cross-interaction module and an inter-scale feature FM, respectively, that enforce communication of different feature maps at the same scale and integrate feature maps of different scales. Notably, both feature extraction and feature fusion of bitemporal RS images can be implemented in a parallelized manner. Finally, a certain conventional change prediction method is applied to the aggregated feature maps, realizing the CD of RS images in different time series.

A. Multiscale Global and Local Feature Mining

Inspired by the wholesome competition between convolutional networks that can extract local information and Transformers that can harvest global feature representation, we propose to apply these two subnetworks in parallel to obtain multiscale local and global feature maps, respectively. In particular, we employ the widely circulated ResNet18 [47] in the CNN branch and adopt the latest PVTv2-B1 [48] in the Transformer branch, individually obtaining the multiscale feature maps. For each network, our implementation consists of three scales with downsampling factors 4, 8, and 16. On the

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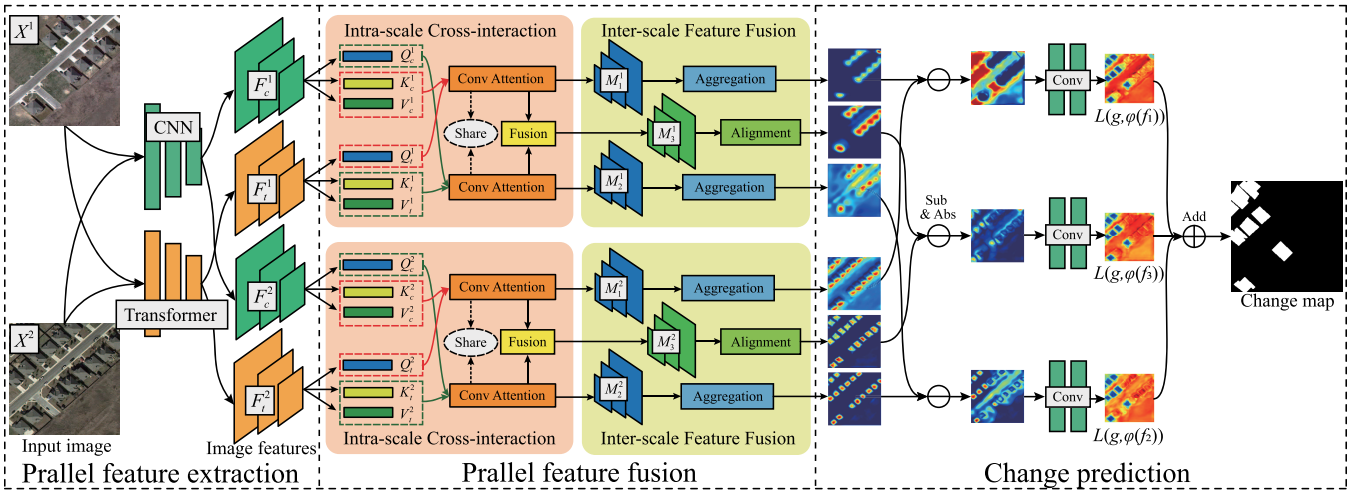

Fig. 1. Architecture of the proposed ICIF-Net. For parallel feature extraction, dual branches of CNN and Transformer are constructed to capture multiscale local features and global features, respectively. For parallel feature fusion, the intra-scale cross-interaction module enabling two-branch communication at the same resolution and the inter-scale feature FM enabling effective integration of multiscale feature maps are proposed. Afterward, a shallow CNN is applied to generate pixel-level change predictions for each branch, followed by an additive operation to generate the final output.

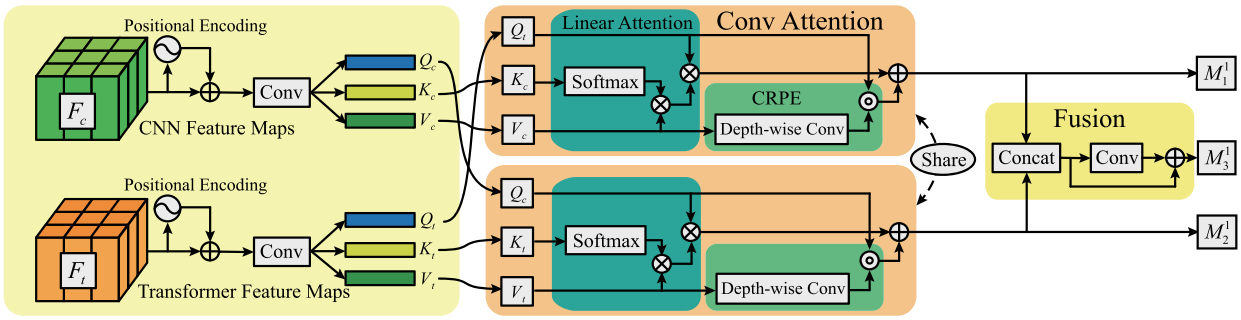

Fig. 2. Illustration of intra-scale cross-interaction. The feature maps from CNN and Transformer are first linearly expanded into vectors. Subsequently, the CA computation is performed through the same linearized Conv Attention module. Finally, the interactive multiscale feature maps are captured directly or by the FM.

one hand, benefiting from the local perception and parameter sharing mechanisms, ResNet18 is dedicated to gradually expand the receptive field for fine-grained information. On the other hand, PVTv2-B1, as a representative variant of pyramid vision transformer (PVT), can guarantee a powerful and reliable performance on generating multiscale features with long-distance dependencies. Functionally speaking, among these two, local features focus on fine-grained details for tiny-object detection and pixel-level localization, while global features focus on the mining of target-level relationships between long-distance components. Taking the ith RS image X^i input to the asymmetric dual-stream encoder as an example, the multiscale local features F_c^i are acquired through the CNN branch, and the multiscale global features F_t^i are generated through the Transformer subnetwork.

B. Intra-Scale Cross-Interaction

Aiming at the core issue of how the feature maps extracted from different backbones communicate within an equivalent scale, as shown in Fig. 2, we propose an intra-scale cross-interaction module to capture the complementary information of the identical image in two different perspectives. Motivated by the cross-attention (CA) mechanism, we propose to develop the linearized Conv Attention method for interactive propagation, which motivates one branch to glimpse the representations of the other while holding the original features. Note that although the generated feature maps from the dual branches are naturally in tensor mode, the CA computations require a sequence of vectors as input. Given the spatially unified-scale features $F_c \in R^{H\times W\times C_1}$ and $F_t \in R^{H\times W\times C_2}$, each feature map should be projected into three intermediates, i.e., query Q, key K, and value V, using three learned linear transformations W^Q, W^K, and W^V, respectively. In our implementation, we perform parallel operations to jointly process F_c and F_t. In particular, the conventional positional encoding operation to supplement the position information and the 1×1 convolution operation to unify and reduce the channel dimension are successively performed. As for the enforcement of information interaction between F_c and F_t, the linearly unfolded Qs from CNN and Transformer are alternatively used to fulfill feature coupling. For simplicity, the same self-attention mechanism module is employed. Note that the conventional operation performs a scaled dot-product operation as formulated in

$$\text{Self Att}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{1}$$

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where T represents the transpose operation, the input vector sequences $Q, K, V \in R^{N\times d}$, $N = H \times W$, and d is the reduced dimension.

However, the complexity of performing a single self-attention operation approximates $O(N^2)$ space complexity and $O(N^2 d)$ $(d \ll N)$ time complexity, which will result in huge resource consumption. Inspired by recent works [49], [50] on linearization of self-attention, we factorize the attention map and compute the second matrix multiplication (key K and value V) together for linear attention, achieving interactive propagation between the feature maps of the CNN branch and the Transformer branch at the same spatial scale. The detailed operation is given in

$$\text{Linear Att}(Q, K, V) = \frac{Q}{\sqrt{d}}\left(\text{softmax}(K)^T V\right) \tag{2}$$

where softmax$(\cdot)$ is employed across the tokens in the sequence in an element-wise manner. Linear attention approaches require $O(Nd)$ space complexity and $O(Nd^2)$ time complexity, which is a significant reduction in capacity and computational cost. It is worth mentioning that although factorization of attention is not an unbiased approximation of scaled dot-product attention, it is still considered as a generalized attention mechanism using the query, key, and value vectors to model feature interactions.

However, the second matrix multiplication (key K and value V) is first calculated in linear attention, which can be regarded as a global data-dependent linear transformation for query Q. Consider the hypothesis that if there are two identical query vectors $q_1, q_2 \in R^d$ from Q, the output of their corresponding attention will be the same. That is to say, features cannot interactively communicate using only linear layers and attention mechanisms. To tackle this issue, convolutional relative position encoding (CRPE) is also introduced into the Conv Attention module, making the output of a token affected both by the query and the locally nearby features. Concretely, as shown in (3), we take the special case of depthwise convolution, as a relative position encoding, to model the local position-based relationships between Q and V

$$\text{CRPE}(Q, V) = Q \circ \text{DepthwiseConv}(V) \tag{3}$$

where $\circ$ is the Hadamard product. Therefore, as shown in Fig. 2, the complete linearized Conv Attention module includes two submodules, i.e., linear attention and CPRE, to complete the interactive communication of different branches and generate multiscale interactive feature maps M_1 and M_2. The complete calculation expression is as follows:

$$\begin{aligned}\text{Conv Att}(Q, K, V) &= \text{Linear Att}(Q, K, V) + \text{CRPE}(Q, V) \\ &= \frac{Q}{\sqrt{d}}\left(\text{softmax}(K)^T V\right) \\ &\quad + Q \circ \text{DepthwiseConv}(V).\end{aligned} \tag{4}$$

Note that for cross-interaction, we only apply Q of one branch to connect with K, V of the other branch, which inevitably results in inconsistent focus due to the difference in feature space between two branches. Therefore, as shown in Fig. 2, aiming at suppressing the feature differences, we employ a Conv Attention module to strengthen interaction efficacy. In addition, we propose the FM to further aggregate the two outputs M_1 and M_2 after cross-interaction, to pursue the mining of praiseworthy features and discard the interference of irrelevant features. In particular, we only perform a Concat operation followed by a residual convolution to acquire the integrated feature map M_3.

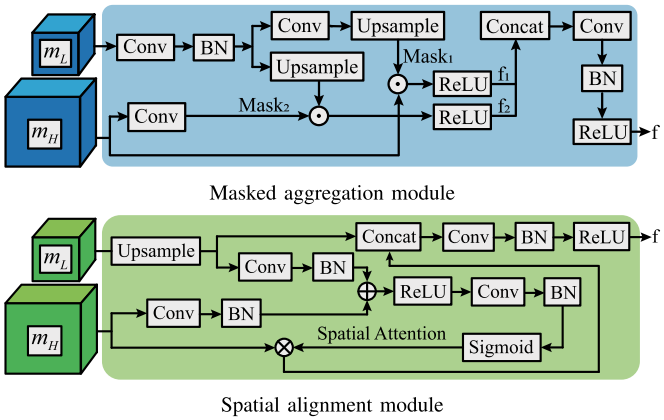

Fig. 3. Illustration of two inter-scale feature FMs.

C. Inter-Scale Feature Fusion

For CD applications, multiscale feature maps enjoy a high reputation in satisfying the target localization needs as well as supplying sufficiently detailed information. However, it is challenging to fuse multiscale feature maps to pursue comprehensive and discriminative output. To highlight the respective properties of interest in reconstructed features M_1 and M_2, as illustrated in Fig. 3, we introduce a masked aggregation (MA) module for feature decoding in dual branches. In particular, as given in (5)–(9), mask maps Mask_1 and Mask_2 are generated by convolving one feature map to another in the form of Hadamard product, suppressing ambiguity by the irrelevant features while activating the mostly task-relevant ones

$$\text{Mask}_1 = \text{Up}(\text{Conv}(\text{BN}(\text{Conv}(m_L)))) \tag{5}$$

$$\text{Mask}_2 = \text{Conv}(m_H) \tag{6}$$

$$f_1 = \text{ReLU}(\text{Mask}_1 \circ m_H) \tag{7}$$

$$f_2 = \text{ReLU}(\text{Maks}_2 \circ \text{Up}(\text{BN}(\text{Conv}(m_L)))) \tag{8}$$

$$f = \text{ReLU}(\text{BN}(\text{Conv}(\text{Concat}(f_1, f_2)))) \tag{9}$$

where m_L and m_H are the low-scale feature map and the high-scale feature map, respectively, Conv, BN, ReLU, Up, and Concat represent the convolution operation, batch normalization, activation function, upsampling method, and merging along channels, respectively. In addition, mask_1 obtained from the low-scale feature map m_L is applied to the high-scale feature map m_H to obtain f_1, focusing on collecting features, such as boundaries and spatial structure. However, the learned features may be sensitive to background noise. As a supplement, Mask_2 obtained from the high scale feature map m_H introduces rich semantic information into m_L, assisting in locating and suppressing noise for f_2. As a result, the output f incorporates multiple meritorious characteristics in multiscale features.

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As for M_3 developed by fusing M_1 with local characteristics and M_2 with long-distance dependencies, we introduce a spatial alignment (SA) module based on the attention mechanism. Hammering at the shared information related both to the local and global features, we model the overall spatial attention distribution by combining different scale features m_L and m_H

$$f_{\text{merge}} = \text{BN}(\text{Conv}(\text{Up}(m_L))) + \text{BN}(\text{Conv}(m_H)) \quad (10)$$

$$\alpha = \text{Sigmoid}(\text{BN}(\text{Conv}(\text{ReLU}(f_{\text{merge}})))) \quad (11)$$

$$f = \text{ReLU}(\text{BN}(\text{Conv}(\text{Concat}(\text{Up}(m_L), \alpha \cdot m_H)))) \quad (12)$$

where Sigmoid is the activation function that maps the values to the interval (0,1), and α represents the attention coefficient. In α, the values of the target regions will tend to be one while the background regions would tend to be zero, being responsible for capturing the task-related features and suppressing the interference of irrelevant information.

D. Prediction Head and Loss Function

In the prediction phase, as shown in Fig. 1, we mainly transplant the CD head previously employed in BIT [29]. In particular, the difference maps between the bitemporal images are computed by subtraction and absolute operations; then, the channels are changed to elements of foreground and background by two 3×3 convolution operations with BatchNorm. Note that in contrast to BIT that contains merely one prediction output, we capture three outputs $f_1, f_2, f_3 \in R^{H_0 \times W_0 \times 2}$ (H_0 and W_0 are the height and width of the feature maps, respectively) focusing on local and global characteristics. In the training phase, we optimize the model parameters by minimizing the cross-entropy loss between f_1, f_2, f_3 and the ground truth (g) one by one. The total training loss is as follows:

$$L_{\text{total}} = \lambda_1 L(g, \phi(f_1)) + \lambda_2 L(g, \phi(f_2)) + \lambda_3 L(g, \phi(f_3)) \quad (13)$$

where ϕ includes the sigmoid function pixel-wisely operated along the channel dimension and an upsampling operation that is uniformly consistent with the size of the input images, and λ_1, λ_2, and λ_3 are tradeoff parameters used to regulate the influence of each loss. In the inference stage, the prediction mask is calculated by performing a pixel-by-pixel Argmax operation on the channel dimension of the sum of $\phi(f_1), \phi(f_2), \phi(f_3)$.

IV. Experiments

A. Data Description

To verify the practical performance of our ICIF-Net, we conduct extensive experiments on four representative bitemporal RS image CD databases.

1) LEarning, VIsion, and RS [52] and its recently expanded version (LEVIR-CD+) are publicly available datasets on large-scale building CD, which, respectively, contain 637 and 985 pairs of RS images with a spatial resolution of 0.5 m/pixel and a total 1024×1024 pixels. Due to the limitation of GPU memory capacity, we simply follow the original sample division of the dataset and uniformly cut the image into 256×256 patches without consideration of the overlapping mechanism. In particular, LEVIR-CD+ uses 10 192 and 5 568 pairs of patches for training and testing, respectively.
2) Wuhan University (WHU-CD) [1] is a building CD dataset containing a pair of 0.2 m/pixel aerial images with a spatial size of $32\,507 \times 15\,354$, which records building changes in the region of Christchurch, New Zealand, after the 2011 magnitude 6.3 earthquake. As [1] provides little analysis of the sample division solution, we cut the large-scale image pairs into 256×256 patch pairs and randomly divide them into 6096/762/762 for training/verification/test.
3) Guangzhou Dataset (GZ-CD) [53] is a high-resolution satellite image CD dataset with a spatial resolution of 0.55 m/pixel, which covers the suburb areas of Guangzhou, China, from 2006 to 2019. Through the Google Earth service, a total of 19 image pairs that vary with the seasons from 1006×1168 pixels to 4936×5224 pixels are collected. In consistent with the processing of the previous two datasets, we uniformly cut the images into the size with 256×256. The total number of samples is 2504/313/313 for training/verification/test, which is a comparatively small-sized dataset compared with LEVIR-CD+ and WHU-CD.
4) Sun Yat-sen University Dataset (SYSU-CD) [31] is a recently developed large-scale CD dataset, whose images are taken in Hong Kong. It contains 20 000 orthographic aerial pairs with 256×256 pixels and a spatial resolution of 0.5 m. Among them, a total of 8000 image pairs are divided equally into validation and test sets, and the remainder serves as the training part. Notably, the SYSU-CD dataset provides multiple change types in relatively complex scenarios, leading to a challenging CD mission.

B. Experimental Setup

1) Comparison to SOTA: To demonstrate the effectiveness of our ICIF-Net on the CD task of bitemporal RS images, several SOTA models are selected as the competitors, including four purely convolution-based methods, i.e., fully convolutional early fusion network FC-EF [22], fully convolutional Siamese-difference network FC-Siam-Di [22], fully convolutional Siamese-concatenation network FC-Siam-Conc [22], and fully residual network FresUNet [51], three attention-based methods, i.e., deeply supervised image fusion network IFNet [25], dual task constrained deep Siamese convolutional network DTCDSCN [32], and deep supervised image fusion network SNUNet [21], as well as the BIT [29] model that tandems CNN and Transformer in a sequential manner.

2) Implementation Details: Our model is built using the PyTorch framework and trained on two NVIDIA TITAN RTX GPUs, each with 12 GB of video memory. In consistent with other comparison experiments, we perform common data enhancement on the input image patches, including flipping, rescaling, cropping, and Gaussian blurring. To update the network parameters, we utilize the stochastic gradient descent (SGD) algorithm. The batch size, the learning rate,

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TABLE I

COMPARISON RESULTS ON DETECTION OF BUILDING CHANGES. THE BEST SCORES ARE MARKED IN BOLD. ALL RESULTS ARE DESCRIBED AS PERCENTAGES (%)

	LEVIR-CD+ Pre. / Rec. / F1 / Iou / OA	WHU-CD Pre. / Rec. / F1 / Iou / OA	GZ-CD Pre. / Rec. / F1 / Iou / OA
FC-EF [22]	70.60 / 73.64 / 72.09 / 56.36 / 97.68	78.79 / 68.15 / 73.08 / 57.58 / 97.61	81.29 / 68.61 / 74.41 / 59.25 / 95.74
FC-Siam-Di [22]	79.94 / 72.55 / 76.06 / 61.37 / 98.14	80.44 / 54.59 / 65.04 / 48.19 / 97.20	80.36 / 61.80 / 69.87 / 53.69 / 95.19
FC-Siam-Conc [22]	78.07 / 68.22 / 72.82 / 57.25 / 97.92	75.89 / 69.30 / 72.61 / 56.99 / 97.51	80.37 / 68.97 / 74.23 / 59.03 / 95.68
FresUNet [51]	78.62 / 79.30 / 78.96 / 65.24 / 98.28	86.55 / 77.68 / 81.88 / 69.32 / 98.36	87.97 / 77.44 / 82.48 / 70.65 / 97.10
IFNet [25]	83.77 / 80.32 / 82.29 / 70.97 / 98.61	92.24 / 75.78 / 83.11 / 71.12 / 98.59	87.39 / 64.57 / 77.65 / 63.47 / 96.64
DTCDSCN [32]	83.41 / 79.08 / 81.19 / 68.76 / 98.58	78.65 / 81.24 / 79.92 / 66.56 / 98.05	86.08 / 73.86 / 79.51 / 66.01 / 96.56
SNUNet [21]	80.52 / 79.43 / 80.82 / 68.17 / 98.42	85.58 / 80.99 / 83.22 / 71.26 / 98.44	85.81 / 79.63 / 83.32 / 71.88 / 97.19
BIT [29]	83.62 / 79.01 / 81.25 / 68.42 / 98.51	86.64 / 81.39 / 83.98 / 72.39 / 98.52	86.00 / 75.85 / 80.61 / 67.52 / 96.71
Our ICIF-Net	**87.79 / 80.88 / 83.65 / 71.89 / 98.73**	**92.98 / 85.56 / 88.32 / 79.24 / 98.96**	**90.17 / 81.57 / 85.65 / 74.91 / 97.53**

the momentum, the weight decay, and the number of training epochs are set to 8, 1e-2, 0.99, 5e-4, and 200, respectively.

3) Evaluation Metrics: For the performance evaluation, we use five metrics to measure the similarity between the predicted change probability maps and the ground truth, including Precision (Pre.), Recall (Rel.), F1-score (F1), Intersection over Union (IOU), and overall accuracy (OA). Each metric is defined as follows:

$$\text{Precision} = \text{TP}/(\text{TP}+\text{FP}) \tag{14}$$

$$\text{Recall} = \text{TP}/(\text{TP}+\text{FN}) \tag{15}$$

$$\text{F1} = 2/(\text{Recall}^{-1}+\text{Precision}^{-1}) \tag{16}$$

$$\text{IOU} = \text{TP}/(\text{TP}+\text{FP}+\text{FN}) \tag{17}$$

$$\text{OA} = (\text{TP}+\text{TN})/(\text{TP}+\text{TN}+\text{FP}+\text{FN}) \tag{18}$$

where TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative, respectively. Notably, F1 is a composite metric that integrates both Precision and Recall.

C. Performance Comparison

Aiming at assessing the effectiveness of the proposed ICIF-Net model, extensive experimental results conducted on three bitemporal RS images CD datasets are quantitatively summarized. Table I reports the overall performance on the test samples of LEVIR-CD+, WHU-CD, and GZ-CD, in which the quantitative results show that our ICIF-Net consistently outperforms the other competitors across the five metrics with a significant margin. For instance, the F1-score of our ICIF-Net surpasses the latest BIT model by 2.4%/4.3%/5.0% on the LEVIR-CD+/WHU-CD/GZ-CD dataset, respectively. Objectively speaking, our ICIF-Net achieves breakthroughs in all five indicators on three datasets, which certificates that introducing intra-scale cross-interaction and inter-scale feature fusion to design a novel network architecture is worthwhile and constructive for CD tasks. In addition, the visualization results of each model on various datasets are shown in Figs. 4–6. Notably, diverse colors are assigned to distinguish the correctness of the detection on different regions, including TP (white), TN (black), FP (red), and FN (green).

1) Visualization on LEVIR-CD+: Notably, LEVIR-CD+ has approximately 80 000 building examples with a time span from 5 to 14 years, covering villas, high-rise apartments, small garages, and large warehouses. As displayed in Fig. 4, we have selected typical samples for visual comparisons, such as small-sized buildings at the edge of Fig. 4(a), the densely changing buildings in Fig. 4(b), the strong seasonal and lighting changes in Fig. 4(c) and (d), and the large scene changes in Fig. 4(e) and (f). The changed region in Fig. 4(a) is relatively easier to detect, and as a result, FC-Siam-Di, FC-Siam-Conc, DTCDSCN, SNUNet, and our ICIF-Net perform slightly better. Yet, the others still suffer from the response of missed detection. In Fig. 4(b), compared with BIT, the results of SNUNet and our ICIF-Net evidently have less missed detections and FP ones of adhesion between buildings. As for the interference of seasonal changes and light-producing shadows shown in Fig. 4(c) and (d), our ICIF-Net shows a clean sweep relative to the competitors. For Fig. 4(e) and (f), fewer false and missed detections verify the capability of our ICIF-Net on detecting changes in scenes with huge variations. In a nutshell, the qualitative results of ICIF-Net on LEVIR-CD+ reach SOTA performance, which is consistent with the quantitative results in Table I.

2) Visualization on WHU-CD: In a similar way to the selection of samples in LEVIR-CD+, as shown in Fig. 5, we pick representative samples from WHU-CD to facilitate visual performance comparison, including small-sized buildings located at the margin of Fig. 5(a), buildings with different shooting angles in Fig. 5(b) and (c), and the serious interference generated by the roads covered with cement or asphalt in Fig. 5(d)–(f). The challenge point in Fig. 5(a)–(d) is to alleviate missed identifications, where SNUNet and our ICIF-Net exhibit more accurate building recognition in visual presentation than the other competitors do. Moreover, under the interference of the roads in Fig. 5(e) and (f), only our ICIF-Net can still maintain high identification and present low false alarms.

3) Visualization on GZ-CD: Witnessing the urbanization process in the suburbs of Guangzhou, China, the representative samples in the GZ-CD dataset are presented in Fig. 6. Representatively, in Fig. 6(b), high-rise buildings are exposed to light and suffer from the negative shading of the nearby ones. In Fig. 6(c), while the arrayed buildings are actually unchanged, the seasonal change significantly hampers the identification accuracy. In Fig. 6(e), large-scale factories are

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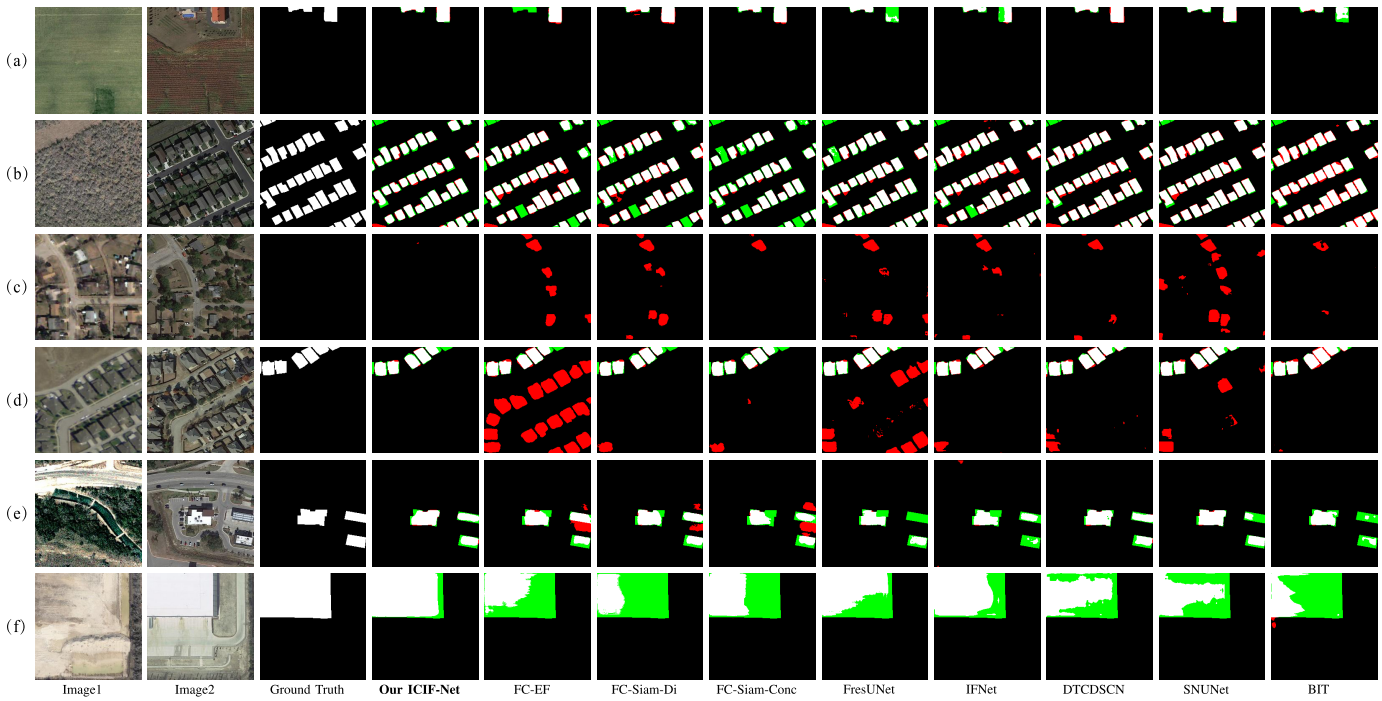

Fig. 4. Visual results of different models on test samples of LEVIR-CD+. (a)–(f) Six representative samples. Different colors are used to facilitate a better observation. White represents a TP, black is a TN, red indicates an FP, and green stands as an FN.

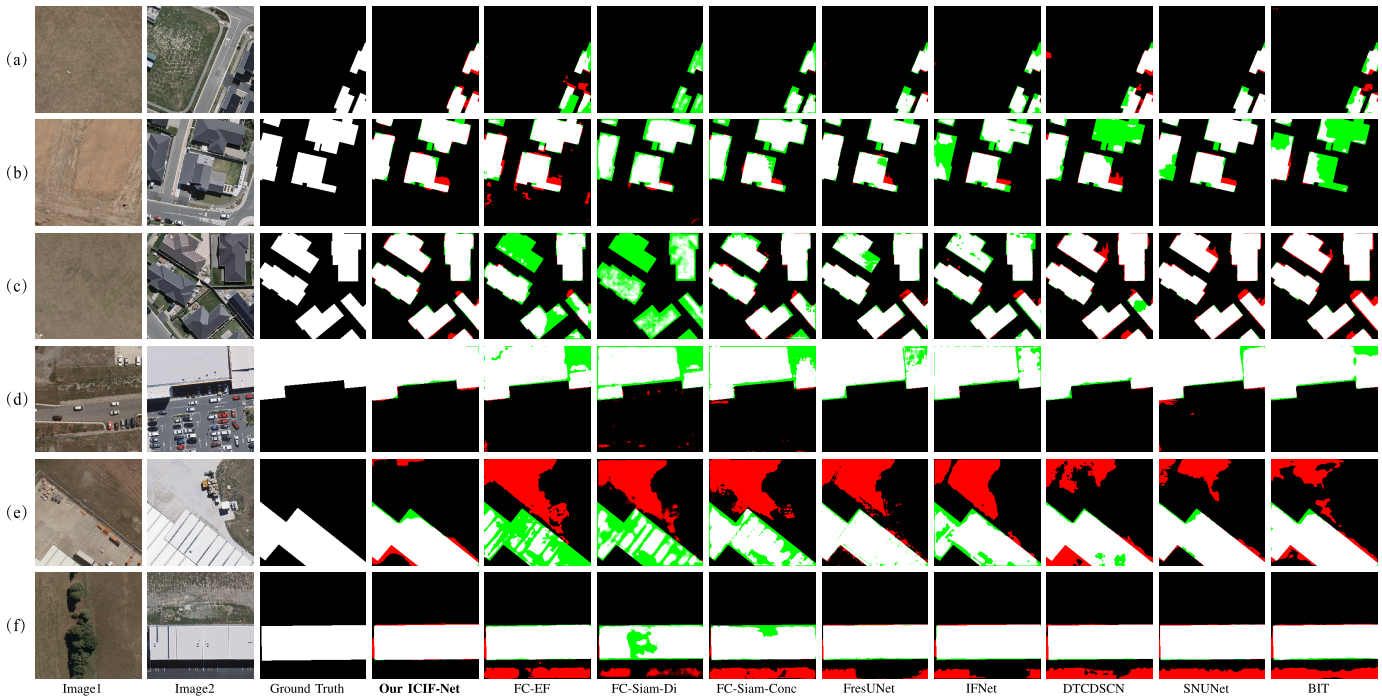

Fig. 5. Visual results of different models on test samples of WHU-CD. (a)–(f) Six representative samples. Different colors are used to facilitate a better observation. White represents a TP, black is a TN, red indicates an FP, and green stands as an FN.

covered by the surrounding plants. Encouragingly, while limited samples are available, our ICIF-Net owns the best visual performance, confirming closely to the quantitative results in Table I. Considering all three datasets, ICIF-Net consistently yields the highest identification scores with the minimum FPs and misses over other competitors, demonstrating its serviceability and superiority for the CD task of RS images.

4) Visualization on SYSU-CD: Note that the other three datasets mainly focus on the changes of different buildings. To further validate the detection performance on various types of changes, as shown in Table II and Fig. 7, we conduct experiments on the SYSU-CD dataset. Quantitative and qualitative experimental results demonstrate that our ICIF-Net performs more accurately with fewer

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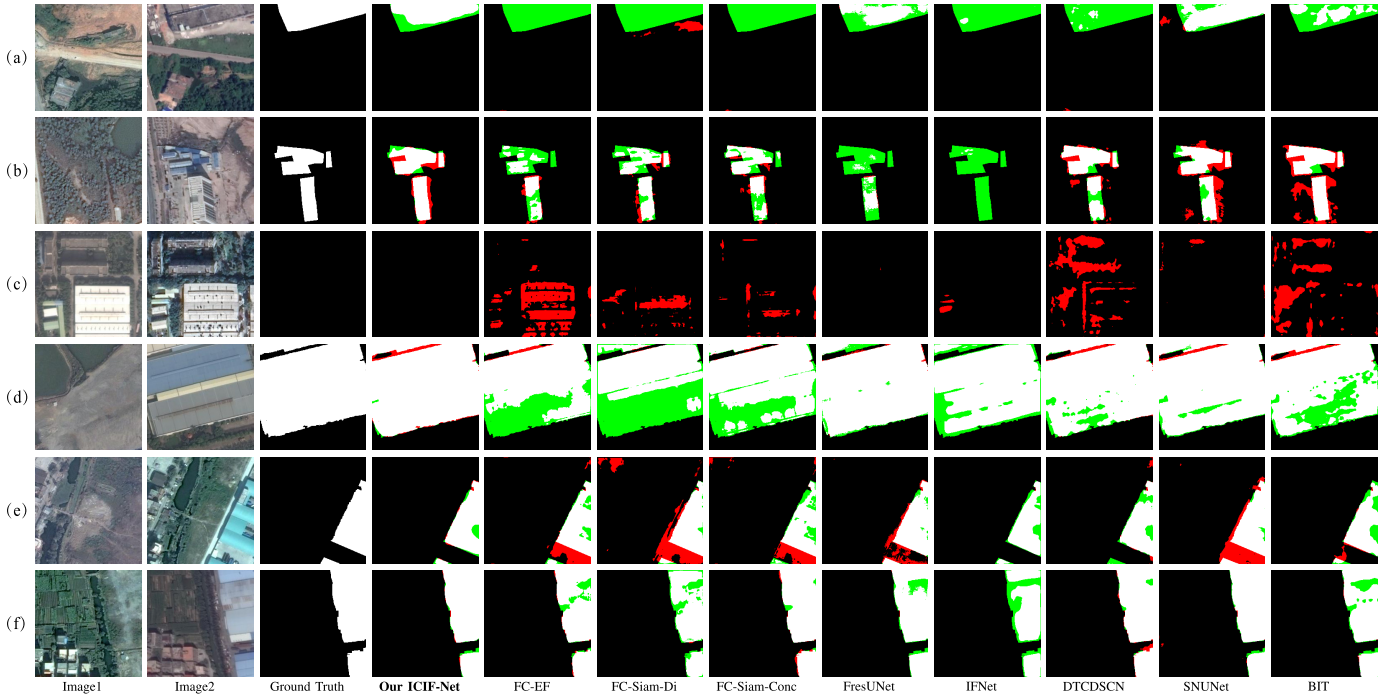

Fig. 6. Visual results of different models on test samples of GZ-CD. (a)–(f) Six representative samples. Different colors are used to facilitate a better observation. White represents a TP, black is a TN, red indicates an FP, and green stands as an FN.

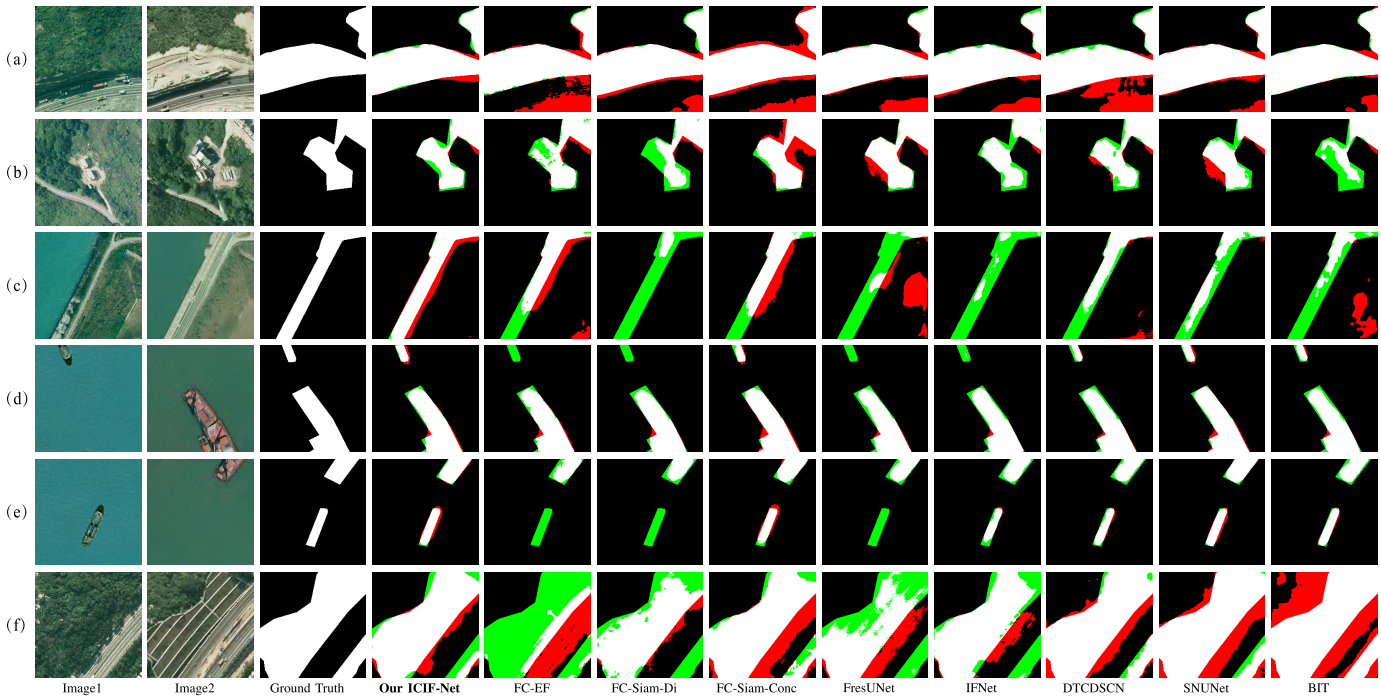

Fig. 7. Visual results of different models on test samples of SYSU-CD. (a)–(f) Six representative samples. Different colors are used to facilitate a better observation. White represents a TP, black is a TN, red indicates an FP, and green stands as an FN.

false and missed pixels in the detection regions, such as roads [Fig. 7(a) and (b)], embankments [Fig. 7(c)], vessels [Fig. 7(d) and (e)], grounds [Fig. 7(f)], as well as changed buildings, over other competitors.

As for the comparison of model efficiency, Table III exhibits the number of model parameters (Params.), the floating-point operations per second (FLOPs), and the cost time needed to run an epoch on GZ-CD dataset, for which the size of used images is $256 \times 256 \times 3$. Due to the parallelized implementation of feature extraction and feature fusion, our ICIF-Net enjoys better semantic representation with the sacrifice of more resource demands. The number of parameters in our model is

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TABLE II
COMPARISON RESULTS ON MULTIPLE CHANGE TYPES

	SYSU-CD Pre. / Rec. / F1 / Iou / OA
FC-EF [22]	72.28 / 72.85 / 72.57 / 56.94 / 87.01
FC-Siam-Di [22]	**84.94** / 51.45 / 64.08 / 47.15 / 86.40
FC-Siam-Conc [22]	83.03 / 71.62 / 76.89 / 62.45 / 89.35
FresUNet [51]	78.56 / 74.96 / 76.38 / 61.85 / 89.12
IFNet [25]	79.59 / 73.58 / 76.53 / 61.91 / 89.17
DTCDSCN [32]	83.19 / 77.25 / 80.11 / 66.82 / 90.96
SNUNet [21]	83.49 / 76.37 / 79.77 / 66.35 / 90.87
BIT [29]	77.69 / 74.53 / 76.08 / 61.39 / 88.95
Our ICIFNet	83.37 / **78.51** / **80.74** / **68.12** / **91.24**

TABLE III
COMPARISON OF MODEL EFFICIENCY. WE REPORT THE NUMBER OF PARAMETERS (PARAMS.), FLOPS, AND THE TRAINING TIMES FOR A SINGLE EPOCH ON GZ-CD DATASET. THE SIZE OF IMAGES USED TO CALCULATE THE FLOPS IS $256 \times 256 \times 3$

Model	Params.(M)	FLOPs(G)	Epoch Time(Min)
FresUNet [51]	1.11	2.02	0.5
IFNet [25]	50.71	82.26	2.3
DTCDSCN [32]	41.07	20.44	1.3
SNUNet [21]	12.04	54.82	1.3
BIT [29]	3.55	10.60	1.0
ResNet18	2.78	3.68	-
PVTv2-B1	13.49	5.32	-
Our ICIF-Net	23.82	25.36	1.7

naturally higher than SNUNet, BIT, and FresUNet, while less than DTCDSCN and IFNet. In terms of the FLOPs, the rank of our model also lies between DTCDSCN and SNUNet, yet follows a reversed order. Unfortunately, the training efficiency of ICIF-Net lags behind both DTCDSCN and SNUNet. Note that two pre-trained backbones are introduced as the individual branches. While fewer parameters are required for the ResNet 18 model in CNN branch, the PVTv2-B1 model used in Transformer branch occupies a great number of parameters. Overall, although our ICIF-Net outperforms the recent IFNet in terms of the number of parameters, FLOPs, and training speed, there is still a requirement for investigation of some lightweight architectures and tap the potential for acceleration.

D. Ablation Analysis

Intended to further validate the impact of various modules and distinct branches on the CD performance, as shown in Tables IV and V, we carry out comprehensive ablation experiments in this section.

1) Module Verification: By removing the modules one by one, we verify the effectiveness of each component in ICIF-Net. The experimental results are shown in Table IV, where "w/o" is the abbreviation of "without," "two CA" indicates that the model employs two CA modules, "CA" and "FM" are denoted as cross-attention and fusion modules in intra-scale cross-interaction, and "MF" is a multiscale feature fusion, including MA module and SA module.

TABLE IV
ABLATION STUDY ON DIFFERENT MODULES

	LEVIR-CD+ F1 / Iou	WHU-CD F1 / Iou	GZ-CD F1 / Iou
w/o CA	82.83 / 70.69	86.62 / 76.44	84.39 / 73.01
two CA	82.76 / 70.59	87.81 / 78.10	85.17 / 74.45
w/o FM	82.72 / 70.54	87.26 / 77.83	84.67 / 73.42
w/o MF	82.61 / 70.37	85.95 / 75.36	84.56 / 73.28
all MA	83.47 / 71.63	87.66 / 78.01	85.58 / 74.80
all SA	83.07 / 71.04	88.11 / 78.75	85.51 / 74.69
ICIF-Net	**83.65 / 71.89**	**88.32 / 79.24**	**85.65 / 74.91**

TABLE V
ABLATION STUDY ON DIFFERENT BRANCHES

	LEVIR-CD+ F1 / Iou	WHU-CD F1 / Iou	GZ-CD F1 / Iou
Base_1	78.57 / 66.24	82.14 / 69.69	79.19 / 68.25
SFP_1	81.58 / 69.25	85.94 / 75.35	83.76 / 72.06
Base_2	79.79 / 67.01	83.26 / 71.35	80.21 / 67.57
SFP_2	82.76 / 70.60	86.21 / 75.76	84.27 / 72.83
ICIF-Net	**83.65 / 71.89**	**88.32 / 79.24**	**85.65 / 74.91**

a) Effectiveness of CA: To analyze whether CA supports better performance for the entire network, we apply the self-attention mechanism to replace the CA module. As shown in Table IV, the removal of CA causes evident performance degradation compared with the original ICIF-Net, demonstrating the significance of interactive communication between convolution and Transformer. In addition, to verify that a single CA module is more beneficial to CD task, we further perform experiments using two CA modules. As expected, the quantitative experimental results validate the feasibility and superiority of developing the same Conv Attention module for feature encoding.

b) Effectiveness of FM: Our original intention of designing the FM using intra-scale cross-interaction hammers at narrow the gap between two dual branches of CNN and Transformer, and also to capture a comprehensive representation with both local and global characteristics. As envisioned, the column of Table IV without the FM component gives an unsatisfactory performance, which laterally verifies the guaranteed feature sublimation from the cross-interaction operation.

c) Effectiveness of MF: To investigate the performance of our proposed multiscale feature FMs, we simply replace the MF component with several cascaded upsampling blocks (w/o MF). Evidently, the poorest performance is generated without our fusion scheme. Moreover, the introduction of each submodule, whatever MA or SA, apparently guarantees a promising performance improvement. However, the approach employed in ICIF-Net is more appropriate in contrast to the single adoption of one of them.

2) Branch Verification: Notably, as shown in Fig. 5, our model is functioned by summing the intermediate quantities of different branches and backpropagating them jointly on each branch. Therefore, it is essential to investigate the contribution of the individual branches and their conjunction to the CD task.

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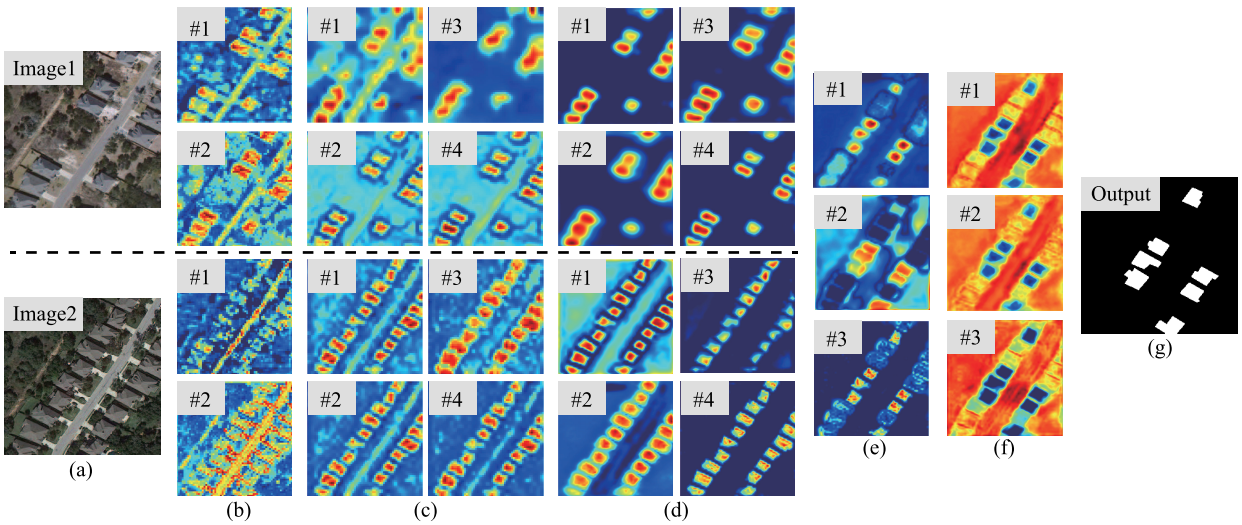

Fig. 8. Network visualization taking bitemporal RS images from LEVIR-CD+ as an example. (a) Input image pairs. (b) Selected high-level feature maps F_c^i (#1) and F_t^i (#2). (c) Selected feature maps M_1^i (#1) and M_2^i (#2) after intra-scale cross-interaction, and M_3^i (#3, #4) obtained by the FM. (d) Selected feature maps A_1^i (#1), A_2^i (#2), and A_3^i (#3, #4) after inter-scale feature fusion. (e) Bitemporal feature difference maps D_1 (#1), D_2 (#2), and D_3 (#3). (f) Change probability maps P_1 (#1), P_2 (#2), and P_3 (#3). (g) Change map.

For this purpose, we simplify the complete ICIF-Net into two baselines (Base_1 and Base_2) and two semifinished products (SFP_1 and SFP_2), where Base_1 adopts the Resnet18 branch alone and is updated to SFP_1 by considering the intra-scale CA of the PVTv2 branch. Similarly, using only PVTv2 without any intervention of convolution, or backpropagating the PVTv2 branch with the support of CNN, can produce Base_2 and SFP_2, respectively. The quantitative results are displayed in Table V, from which we can draw three conclusions. First, benefiting from the superiority of long-range modeling, the Transformer-based models (Base_2 and SFP_2) consistently outperform the CNN-based counterparts (Base_1 and SFP_1). Second, both SFP_1 and SFP_2 are significantly strengthened compared with their respective baselines, authenticating the dominance of intra-scale cross-interaction. Finally, the single-branch models consistently underperform the whole ICIF-Net with all factors considered jointly.

TABLE VI
ABLATION STUDY ON DIFFERENT LOSS FUNCTIONS

$\lambda_1 : \lambda_2 : \lambda_3$	LEVIR-CD+ F1 / Iou	WHU-CD F1 / Iou	GZ-CD F1 / Iou
0 : 0 : 1	83.34 / 71.42	85.32 / 74.39	83.51 / 71.82
1 : 0 : 0	81.58 / 69.25	85.94 / 75.35	83.76 / 72.06
0 : 1 : 0	82.76 / 70.60	86.21 / 75.76	84.27 / 72.83
1 : 0 : 1	83.09 / 71.08	87.38 / 77.03	84.17 / 72.67
0 : 1 : 1	82.38 / 70.11	86.73 / 76.57	84.45 / 73.03
1 : 1 : 0	82.72 / 70.54	86.32 / 75.93	84.67 / 73.42
1 : 1 : 1	83.47 / 71.58	88.24 / 78.96	84.90 / 73.76
3 : 3 : 2	82.79 / 70.73	87.68 / 78.06	85.24 / 74.28
3 : 2 : 2	82.46 / 70.44	88.27 / 79.00	**85.98 / 75.41**
2 : 3 : 2	82.47 / 70.52	85.52 / 74.69	85.28 / 74.34
1 : 1 : 2	83.13 / 71.16	88.13 / 78.73	84.84 / 73.68
1 : 2 : 2	82.94 / 70.98	87.97 / 78.33	84.57 / 73.26
2 : 1 : 2	**83.65 / 71.89**	**88.32 / 79.24**	85.65 / 74.91

E. Parameter Analysis

To assess the practical contribution of different prediction branches and the availability of the fusion results, we conduct ablation experiments on the hyperparameters λ_1, λ_2, and λ_3 of the loss function (13). The resultant metrics are presented in Table VI. Note that the proportions of the three hyperparameters (λ_1:λ_2:λ_3) are taken as the practical parameters. From the first observation, we can see that the overall performance difference is relatively mild against different settings of the tradeoff parameters. Moreover, it can be found that Table VI presents a progressively improved performance from singleton-branch prediction {1:0:0, 0:1:0, 0:0:1} to double-branch prediction {1:1:0, 1:0:1, 0:1:1} and then to triple-branch prediction (1:1:1), which corroborates the effectiveness of merging multibranch outputs. To further investigate the influence of varying parameter ratios on detection performance, we enrich the evaluation by fixing the third branch and altering the weights of the first two branches. The experiments are conducted by strengthening the roles of the first two branches {3:3:2, 3:2:2, 2:3,2} or attenuating the effect of these two {1:1:2, 1:2:2, 2:1:2}, respectively. Under these settings, the performance can reach a higher level on the three CD datasets. Empirically, attenuating the prediction weights of the second branch may be a suitable choice.

F. Network Visualization

Aiming at better comprehending our model, an example is offered to visualize the feature maps at various stages of the ICIF-Net model. Given a bitemporal RS image pair [Fig. 8(a)], the high-level feature maps F_c^i and F_t^i [Fig. 8(b)] are captured by the dual branches of CNN and Transformer. Then, the information interaction between F_c^i and F_t^i is performed by intra-scale cross-interaction to pursue the cross-fusion between

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local features and global features and to obtain M_1^i, M_2^i, and M_3^i [Fig. 8(c)] with robust representation capability. After that, M_1^i, M_2^i, and M_3^i are fed to their respective multiscale aggregation module with inter-scale feature fusion, resulting in A_1^i, A_2^i, and A_3^i [Fig. 8(d)]. Naturally, A_1^i, A_2^i, and A_3^i combine both the low-level detail information and the high-level structural features. From the abovementioned subgraphs, the localization of targets and their contours in each RS image are progressively refined with the depth of the model, which demonstrates the effectiveness of our proposed modeling strategies. Finally, both feature difference maps [Fig. 8(e)] and change probability maps [Fig. 8(f)] accurately exhibit the differences between bitemporal RS images.

V. Conclusion

To deeply aggregate the local and global features, in this article, we shift the design paradigm from series to parallel and propose a CNN-Transformer collaborated model for CD in RS images, namely, ICIF-Net. The intra-scale cross-interaction module revolves around the complementarity between fine-grained local details and target-level global interests, while the inter-scale feature FM is intended to integrate high-level representations that benefit localization and low-level features that facilitate structural sketching. Extensive experiments are conducted on four publicly available datasets, i.e., LEVIR-CD+, WHU-CD, GZ-CD, and SYSU-CD. The experimental results demonstrate that our proposed ICIF-Net achieves SOTA performance compared with most recently published competitors. Due to the introduction of the pre-trained CNN and Transformer backbones in two individual branches, the drawback of our model mainly lies in the massive resource consumption. Therefore, future works will undertake in-depth research on lightweight models. In addition, investigating the availability of our proposal for CD in heterogeneous bitemporal images is also an important work, for example, CD on bitemporal synthetic aperture radar and optical images.

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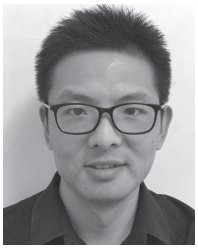

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